Midpoint Progress Report

Out-of-Distribution Generalization in Satellite Land Cover Classification

Course: CSCI 3052U

Team: SixtyFive.AI

Date: October 30, 2026

Team Members: Mohammad Ali Jafari, Rayan Khan, Faraz Shaikh, Haseeb Sajjad, Chris Ibrahim

1. Project Overview and Progress to Date

We are exploring the impact of geolocation on the performance of machine learning algorithms for land cover classification on satellite imagery. We want to explore if a model that was trained on one geolocation performs similarly on satellite images from other geolocations.

We are using the EuroSAT Sentinel-2 dataset which consists of 27,000 labeled satellite images from ten land covers. This is a supervised multiclass classification task as we are given labeled satellite images and we want to classify new satellite images into one of the ten given land covers.

Our primary research question is:

How does the performance of a satellite land-cover classification model change when evaluated on geographically different data compared with development validation data?

Our main evaluation measure will be Macro-F1. This evaluates the F1 score for each class individually and then averages over all the classes. This ensures that all ten classes are treated equally.

Up to the halfway point of the project, we have created a project proposal, prepped the data, conducted exploratory analysis on the data, gathered geospatial information about the data, and performed some initial baseline experiments. For these baseline experiments we employed Random Forest and Principal Component Analysis (PCA). We have found a large drop in performance on out-of-distribution (OOD) data that is geographically split.

For the second half of the project, we plan to continue error analysis, begin studying the proposed ResNet-18 convolutional neural network, refine the evaluation method, check the geographic splits, and so on.

2. Current Methods and Approaches

2.1 Dataset and Data Preparation

We use the EuroSAT v2 dataset, which is composed of images from Sentinel-2 satellites over Europe. The dataset has 27,000 labeled images, and each image can be represented in two ways:

- RGB: $64 \times 64 \times 3$
- Multispectral: $64 \times 64 \times 13$

It should be noted that these are two views of the 27,000 satellite images and not 54,000 separate images.

There are ten land covers in the dataset: Sea/Lake, River, Residential, Industrial, Highway, Forest, Permanent Crop, Pasture, Herbaceous Vegetation, and Annual Crop.

There are 2,000 images in Pasture and 3,000 images in some of the other classes. There is a factor of 1.5 between the largest and smallest class size which we consider to be a fairly well balanced dataset.

In Milestone 2, data validation and exploratory analysis was performed. All 27,000 images in each of the image representations were decoded without issue. No unexpected image sizes, decode errors, non-finite pixel data, or masked/nodata values were detected in the data. Additionally, the data files were confirmed to match their published checksums.

Exploratory analysis included image properties, class distributions, images within each representation, and geographic information extracted from the multispectral GeoTIFFs. Coordinate information was used to create preliminary geographic subdivisions.

Geographic processing is key to our study since we are interested in regional generalization. Some geographic classifications and edge cases will need to be verified prior to final analysis.

2.2 Data Splitting and Preprocessing

Using geographic metadata, we created a development set and an OOD set. The OOD set will be used to test the model on geographically unseen satellite images.

For the base experiments in the development set, we used a 80/20 stratified train/valid split and random seed 42.

For the RGB experiments, the images were downsampled from 64 × 64 to 16 × 16 pixels. This means that each resized image now has 768 numerical RGB pixel features.

PCA was then used to reduce the number of features to 100 before classification.

Data leakage is also important. For the final testing, it is crucial that all data preprocessing transformations are fit only on training data and that geographically correlated samples are split correctly.

2.3 Models Implemented and Planned

Model	**Purpose**	**Configuration**
Majority-class classifier	Establish a simple performance reference	Predicts the majority class
PCA + Random Forest	Classical machine learning baseline	RGB images resized to 16 × 16, PCA with 100 components, Random Forest with 200 trees
ResNet-18	Planned model for learning spatial image features	image features Configuration to be finalized during Milestone 5

A majority class baseline is used to provide a comparison to see if a more sophisticated model actually improves on the results.

Our primary implemented solution is a combination of PCA and Random Forest. PCA is used to transform the image features into a lower dimensional space, and Random Forest, which uses a number of decision trees for satellite image classification.

ResNet-18 will be implemented in the future. Results from ResNet-18 have not been recorded as complete and thus are not included in this report.

2.4 Training and Evaluation Setup

Machine learning implementations are made in Python and we use Jupyter notebooks. Code is stored on GitHub, which allows version control, collaboration and information storage.

For the experiments that have been carried out we have used accuracy and Macro-F1 for classification results.

Accuracy represents the percentage of correct predictions made compared to the actual value. Macro-F1 takes the unweighted mean of the F1 score for each class.

We chose Macro-F1 over accuracy as it treats all ten land covers equally. Some land covers may be harder to classify.

A random seed of 42 has been used for the train set. Further validation using further random splits or spatial groups will be carried out to give a better indication of the reliability of the model.

For the implemented tree based approach there is no use of gradient optimisers or learning rate schedules used in neural networks. This will be relevant when using ResNet-18.

Uncertainty, hardware metrics and training time comparisons are not included in this set of results but will be included if collected in the following experiments.

3. Theoretical Justification for Chosen Methods

3.1 Principal Component Analysis

PCA is a method for dimensionality reduction. It takes our original image features and converts them into a smaller number of principal components. Each principal component is a direction of maximum variance in the image.

We use PCA because satellite images have many numerical pixel features. By reducing the number of features we can provide a more compact input to our classic image classifier.

A problem with PCA is that it only retains directions of maximum variance and does not necessarily learn any image content.

3.2 Decision Trees and Random Forest

A decision tree works by identifying rules in the data which partition the feature space into regions. The rules are applied to partition the data.

A Random Forest is an ensemble of decision trees. The prediction made by a Random Forest is made by combining the predictions of the constituent trees. An ensemble of trees can give more stable predictions and reduce over fitting compared to a single decision tree.

We selected Random Forest because it provides a practical supervised classification baseline and allows us to investigate whether image features extracted through PCA can distinguish different land-cover categories.

This model also connects directly to course concepts such as supervised learning, decision trees, ensemble methods, and overfitting.

3.3 Planned ResNet-18 Model

ResNet-18 is a convolutional neural network that can learn spatial features directly from image data.

Unlike our current PCA and Random Forest baseline, a convolutional neural network is designed to identify patterns such as edges, textures, and increasingly complex spatial structures.

ResNet-18 is therefore a useful model to investigate whether learning image-specific features improves classification performance, particularly on geographically different satellite images.

The model remains part of our next experimental milestone, and no conclusions about its performance can be made yet.

3.4 Generalization and Evaluation Methodology

The central concern of our project is generalization: whether a machine-learning model performs well on new data that differs from the data used during development.

Traditional random validation splits can produce optimistic performance estimates when geographically related satellite images appear in both training and validation datasets.

Our project addresses this concern by comparing development validation performance with a geographically separated OOD evaluation partition.

However, the geographic partitions are currently provisional. More rigorous validation is required to confirm spatial separation and determine whether the observed performance decrease is primarily caused by geographic differences or other factors, such as changes in class distributions.

We use Macro-F1 alongside accuracy because overall accuracy alone may conceal poor performance in individual land-cover categories.

3.5 Alternative Approaches

Alternative	Current Consideration
Majority-class classifier	Implemented as a naive baseline, but not suitable as the main model because it cannot learn meaningful land-cover distinctions
Single decision tree	Not selected as the main classical baseline because Random Forest combines multiple trees and generally offers better prediction stability
ResNet-18	Planned for comparison because convolutional networks can learn spatial image features

4. Preliminary Results and Findings

4.1 Quantitative Results

The initial experiments produced the following results.

Model	Evaluation Partition	Accuracy	Macro-F1
Majority-class	Development validation (n = 3,516)	0.1360	0.0239
Majority-class	Geographic OOD (n = 6,231)	0.0913	0.0167
PCA + Random Forest	Development validation (n = 3,516)	0.7085	0.6910
PCA + Random Forest	Geographic OOD (n = 6,231)	0.4486	0.4365

The majority-class baseline performs poorly on both partitions, as expected from a classifier that predicts only one class.

The PCA and Random Forest model achieves substantially better performance. On development validation data, its accuracy is approximately 70.85%, with a Macro-F1 score of 0.6910.

However, on the geographic OOD test set, accuracy decreases to approximately 44.86%, while Macro-F1 decreases to 0.4365.

The absolute decrease in Macro-F1 is approximately **0.2546 points**, representing a relative reduction of about 36.8% from the development score.

These results suggest that the model has more difficulty classifying satellite imagery from geographically different regions than images from the development distribution.

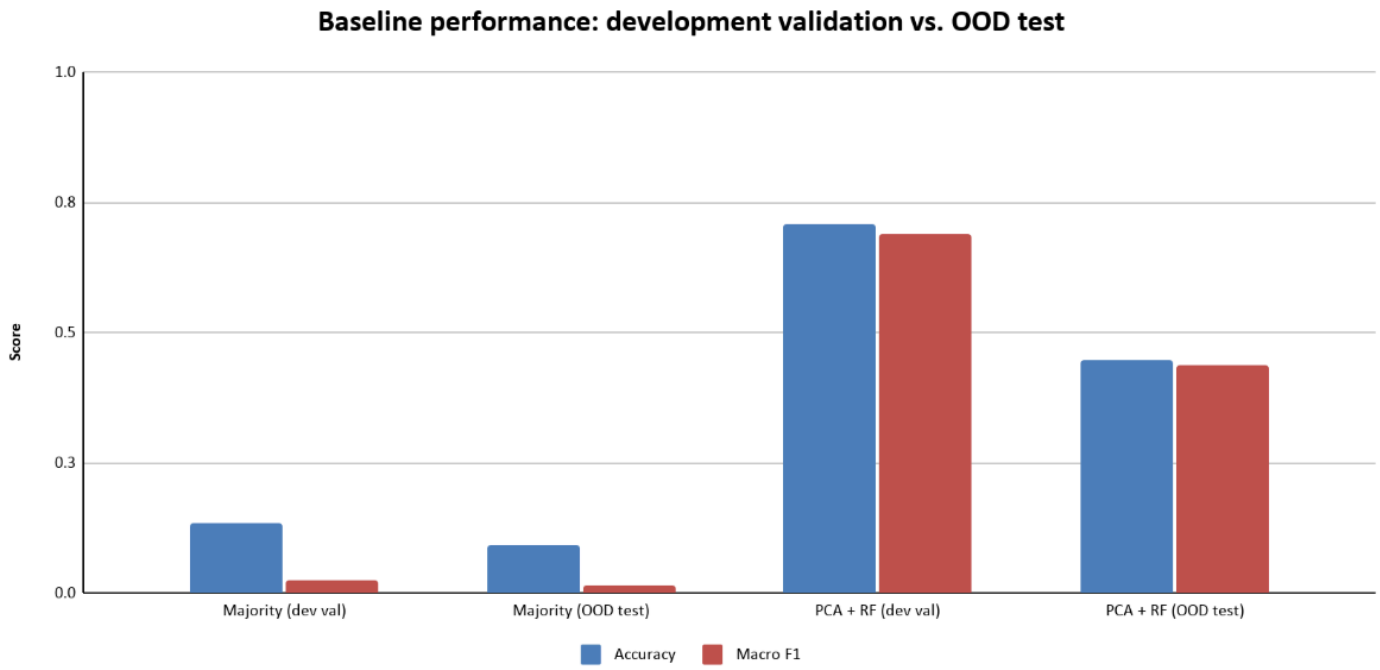
However, the results remain preliminary and do not establish geographic differences as the sole cause of the performance decrease.

4.2 Baseline Performance Visualization

The baseline performance comparison graph illustrates the accuracy and Macro-F1 scores for both models across the development and OOD partitions.

The graph shows that PCA and Random Forest outperform the majority-class baseline in both evaluation settings. However, both accuracy and Macro-F1 decrease on the geographic OOD partition.

Figure 1: Baseline comparison of development validation and geographic OOD performance.



4.3 Per-Class Performance Analysis

We also examined the individual F1 scores of land-cover categories to determine whether all classes experienced similar performance decreases.

Selected results are shown below.

Land-Cover Class	Development F1	OOD F1	F1 Decrease
Pasture	0.772	0.189	0.583
Herbaceous Vegetation	0.729	0.270	0.459
Annual Crop	0.690	0.338	0.352
Residential	0.709	0.364	0.345
Forest	0.917	0.864	0.052
Sea/Lake	0.898	0.877	0.022

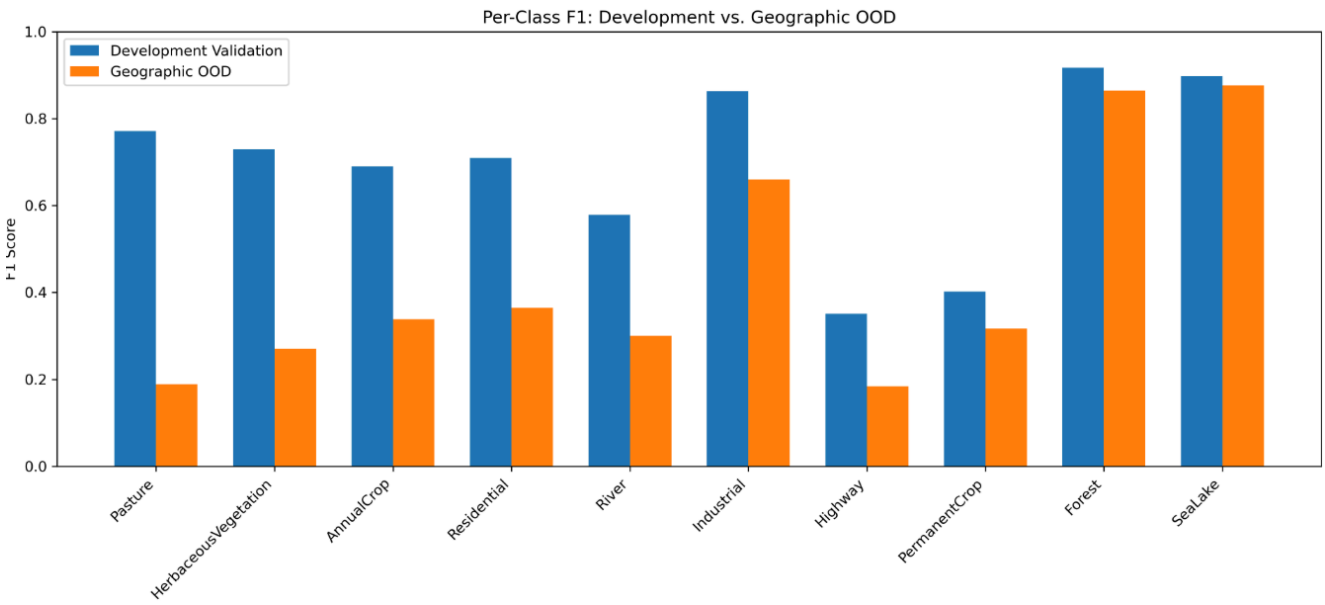
The performance decrease is not uniform across all classes.

Pasture experiences the largest decrease, with F1 falling from approximately 0.772 to 0.189. Herbaceous Vegetation also shows a substantial decrease, from 0.729 to 0.270.

In contrast, Forest and Sea/Lake maintain comparatively strong OOD performance, with F1 scores of approximately 0.864 and 0.877, respectively.

These results suggest that some land-cover categories are more sensitive to differences between the development and OOD partitions.

Figure 2: Comparison of per-class F1 scores across development validation and geographic OOD data



4.4 Error Analysis and Interpretation

Our initial class-level analysis indicates that geographic OOD performance degradation differs considerably across land-cover categories.

One possible explanation is that certain categories, such as Pasture and Herbaceous Vegetation, exhibit more variable textures or regional characteristics. In comparison, other categories may contain more recognizable visual patterns.

However, these explanations remain hypotheses.

We have not yet completed sufficient prediction-level error analysis to identify the precise causes of the differences.

Our next experiments will investigate confusion matrices, class frequencies across geographic regions, and examples of incorrect predictions to better understand the model's failure modes.

4.5 Limitations of Current Results

Our current evaluation has several limitations.

First, the geographic partitions require additional verification, particularly for ambiguous locations and boundary cases.

Second, nearby satellite samples may introduce spatial autocorrelation leakage, potentially affecting the reliability of performance estimates.

Third, differences in class distributions between development and OOD datasets may contribute to the observed performance gap.

Finally, the results do not yet include uncertainty estimates across multiple seeds or geographic validation folds.

For these reasons, we interpret our findings as preliminary evidence of reduced OOD performance rather than a definitive measurement of the effect of geographic shift alone.

5. Deviations from the Original Plan

Our original proposal outlined geographic evaluation, baseline classification, and additional experiments using ResNet-18.

At midpoint, the classical baseline experiments have been completed, while several planned evaluation procedures remain unfinished.

Planned Approach	Current Status	Reason	Impact
Strict geographic evaluation	Provisional geographic partitions created	Some geographic assignments require verification	Current OOD conclusions remain preliminary
Geographic-grouped cross-validation	Not yet completed	Additional geographic validation work required	Performance variability has not been fully established
Random Forest baseline	Completed	Initial classical model for comparison	Provides preliminary performance results
ResNet-18 experiments	Planned for Milestone 5	Classical baseline completed first	CNN comparison not yet available
Detailed error analysis	Initial per-class F1 analysis completed	Further prediction-level investigation required	Specific failure modes remain uncertain

The most significant deviation is that the geographic evaluation process has not yet undergone all planned validation checks.

We intend to address this during Milestone 5 while continuing with the planned CNN comparison.

6. Current Blockers and Challenges

The primary challenges at midpoint involve geographic evaluation reliability, model generalization, and completing the remaining experiments.

Challenge	Effect	Mitigation Strategy	Status
Geographic boundary ambiguity	Some sample assignments require verification	Review ambiguous locations and geographic metadata	Ongoing
Potential spatial leakage	Could produce optimistic evaluation results	Investigate spatial overlap and apply geographic-grouped validation	Planned
Different class distributions	May contribute to performance differences	Compare class frequencies and class-level errors	In progress

Uneven OOD performance	Some classes experience substantial F1 decreases	Conduct confusion-matrix and prediction-level analysis	In progress
remaining ResNet-18 experiments	CNN comparison is not yet available	Prioritize model implementation and evaluation in Milestone 5	Planned

6.1 Mitigation Strategy

Our immediate priority is to improve the reliability of the geographic evaluation procedure.

This includes reviewing geographic assignments, investigating potential spatial leakage, and implementing geographic-grouped validation.

We also plan to expand the class-level analysis to better understand which categories experience the largest performance decreases.

Once the evaluation procedure is sufficiently validated, we will continue with ResNet-18 experiments and compare the results against our completed PCA and Random Forest baseline.

7. Responses to Feedback Received So Far

Feedback source: Team 27

Feedback received: Team 27 reviewed our methodology and preliminary results and recommended improving geographic validation and class-level error analysis.

Our response: We agree that these improvements will strengthen the reliability of our evaluation and provide a better understanding of how geographic differences affect classification performance.

Planned actions: During Milestone 5, we plan to verify geographic partitions, investigate potential spatial leakage, and conduct more detailed class-level error analysis.

Status: Planned for Milestone 5 (November 20, 2026).

8. Revised Timeline and Next Steps

The team is currently following the milestone dates established in the original project proposal.

Phase	Original Date	Revised Date	Status
Milestone 1: Proposal and team planning	September 18, 2026	Unchanged	Completed
Milestone 2: Dataset preparation and EDA	October 2, 2026	Unchanged	Completed
Milestone 3: Baseline reproduction	October 19, 2026	Unchanged	Completed
Milestone 4: Midpoint presentation and report	October 30, 2026	Unchanged	Current milestone
Milestone 5: Final experiments and analysis	November 20, 2026	Unchanged	Upcoming
Milestone 6: Final report and demonstration	December 4, 2026	Unchanged	Upcoming

8.1 Prioritized Next Steps

Priority 1: Geographic Evaluation Validation

Verify geographic assignments, review ambiguous samples, and investigate potential spatial leakage. The goal is to improve confidence in the development and OOD evaluation partitions.

Priority 2: ResNet-18 Experiments

Implement and evaluate the planned ResNet-18 model to compare CNN performance with the current PCA and Random Forest baseline.

Priority 3: Detailed Error Analysis

Expand the current class-level evaluation by investigating confusion matrices, class frequencies, and examples of incorrect predictions.

Priority 4: Final Results and Documentation

Compile the experimental results, document limitations, and prepare the final report and demonstration for December 4, 2026.

8.2 Risks and Contingency Plans

If geographic assignment issues cannot be fully resolved, the team will document the affected samples and any limitations on interpreting the OOD results.

If ResNet-18 experiments encounter computational or implementation difficulties, we will prioritize producing reliable, reproducible evaluations of the completed models rather than presenting incomplete experiments.

If time constraints limit the planned additional analysis, we will focus on the most important findings and clearly identify any remaining limitations in the final report.

9. Midpoint Peer Check-in

9.1 Team Contributions

Our team distributed responsibilities across Milestones 2 and 3, with members collaborating on dataset preparation, model development, and documentation.

For Milestone 2, Mohammad Ali Jafari and Rayan Khan collaborated on dataset preparation, exploratory data analysis, geographic information processing, and project documentation.

For Milestone 3, Faraz, Haseeb, and Chris were the primary contributors to baseline model development and documentation. Mohammad Ali Jafari and Rayan Khan also contributed to Milestone 3, although their contributions were not represented through individual GitHub pushes.

Some work was exchanged directly between team members for integration into the repository. Consequently, the GitHub commit history does not fully represent every member's contribution.

Team Member	**Contribution**
Mohammad Ali Jafari	preparation, EDA, geographic preparation, and documentation, repository integration
Faraz Shaikh	baseline model development and documentation
Haseeb Sajjad	baseline model development and documentation
Chris Ibrahim	baseline model development and documentation
Rayan Khan	preparation, EDA, geographic preparation, and documentation, repository integration

9.2 Team Collaboration

Our team coordinated work through direct communication and shared project files. Tasks were distributed across members, with project work integrated into the shared GitHub repository.

Although technical difficulties affected some direct GitHub contributions, members shared their work to support project completion.

Moving forward, we intend to maintain clearer records of individual task completion and improve the process of integrating contributions into the repository.

9.3 Peer Feedback Exchange

Our team received feedback from Team 27, who reviewed our methodology and preliminary results. They recommended improving geographic validation and conducting more detailed class-level error analysis.

We plan to address these recommendations during Milestone 5 by reviewing the geographic partitions, investigating potential spatial leakage, and expanding our class-level performance evaluation.

Feedback given to Team 27:

Our team reviewed Team 27's NBA player clustering project, which uses K-Means and Gaussian Mixture Models to identify player playing styles. We recommended clarifying how the resulting clusters would be used to identify complementary teammates for a star player. We also suggested investigating whether more than two clusters could produce more meaningful player-style categories and further analyzing cluster stability and interpretability. These recommendations were intended to help connect their machine-learning results more directly to their research objective.

Date: October 29, 2026

10. Conclusion

By the midpoint, Team SixtyFive.AI has completed dataset verification, exploratory data analysis, geographic preparation, and initial baseline experiments for satellite land-cover classification.

Our PCA and Random Forest model achieved a development Macro-F1 score of 0.6910, which decreased to 0.4365 on the geographic OOD test set. This represents a decrease of approximately 0.2546 points and suggests that the model has difficulty generalizing across the current geographic partitions.

However, further geographic split validation and additional experiments are needed before drawing stronger conclusions.

Our remaining work will focus on improving evaluation reliability, completing ResNet-18 experiments, expanding error analysis, and preparing the final project deliverables.

Sources and Reproducibility

Team GitHub Repository:
https://github.com/Rayan-RK/CSCI3052U-Project-TeamSixtyFive

Project Documentation:

- reports/milestone1_proposal.md
- reports/milestone2_approval_memo.md
- reports/ML_phase3_Technical Memo.pdf

Jupyter Notebooks:

- notebooks/01_eda_eurosat.ipynb
- notebooks/02_baseline_reproduction.ipynb

Experimental Results:

- outputs/combined_baseline_results.csv
- outputs/rgb_f1_gap_analysis.csv

These files document the completed data exploration, baseline methodology, preliminary experimental results, and evaluation limitations.

Presentation Link:

https://youtu.be/VPz_xD8i0Zo?si=8TvKxAmrdxiHcpOo